

- 6 A spring with spring constant k is attached to a fixed end as shown in Fig 6.1a. When a mass M is attached to it, they reached an equilibrium when the spring extended by x_0 as shown in Fig 6.1b.

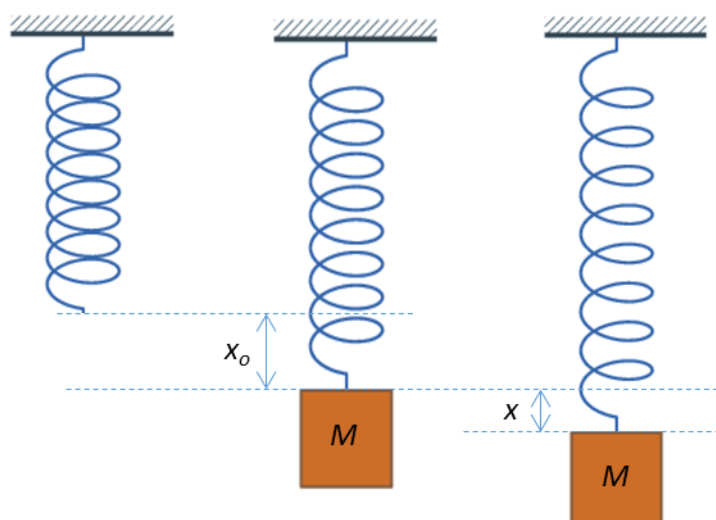


Fig 6.1a

Fig 6.1b

Fig 6.1c

M is now displaced further downwards by a distance x as shown in Fig 6.1c. Mass M is subsequently released.

- (a) Taking downward displacement as positive, show that a , the acceleration of the M and x , it's displacement from the equilibrium position is related by

$$a = -\frac{k}{M}x$$

[2]

- (b) Hence or otherwise, explain if M is oscillating in simple harmonic motion.

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..... [2]

- (c) Given that $k = 150 \text{ N m}^{-1}$ and $M = 2.0 \text{ kg}$, determine f_o , the natural frequency of the spring mass system.

$$f_o = \dots\dots\dots \text{ Hz [2]}$$

- (d) An oscillator is now attached to the same spring mass system as shown in Fig 6.2.

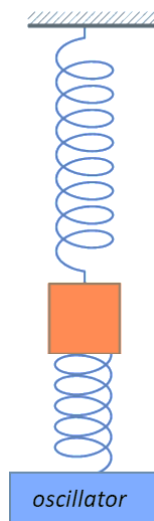


Fig 6.2

The oscillator applies a periodic force of constant magnitude onto the spring mass system.

[Turn over

- (i) State the frequency at which the system will oscillate at when the frequency of the oscillator is set to 5.0 Hz.

frequency = Hz [1]

- (ii) The frequency of the oscillator is varied from 0.0 Hz to 10.0 Hz.

Sketch on Fig 6.3 the variation of amplitude of the oscillation of the spring mass system to the frequency of the oscillator. Air resistance is not negligible.

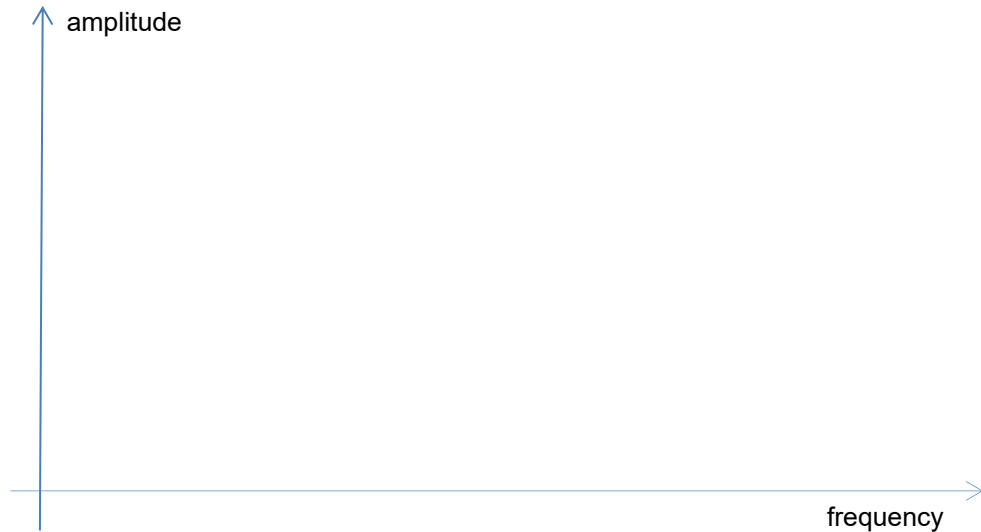


Fig 6.3

[2]

- (iii) Sketch on Fig 6.3 the new graph when the periodic force of the oscillator is increased. Label the graph **F**.

[1]